A Formal Structural Framework for High-Dimensional Representations: Theory and Illustrative Analysis Using SFH-SGP

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Abstract

We introduce SFH-SGP, a formal mathematical framework for characterizing structure in high-dimensional representational systems. The framework defines a set of abstract constructs—Quota, Fertility, and Torsion—which together yield a derived scalar metric, χ (chi), intended to quantify structural organization within a representational space.

We position SFH-SGP as a theory-building contribution rather than a validated empirical model. Accordingly, we (i) formally define the framework, (ii) outline its mapping to potential neural and computational observables, and (iii) provide an illustrative computational analysis using TRIBE v2 outputs as a proof-of-concept.

The analysis demonstrates that χ can induce non-random partitioning of representational states under controlled conditions. However, we do not interpret this as empirical validation. Instead, we propose a set of testable predictions for future work, including expected relationships between χ and neural representational gradients.

This work establishes SFH-SGP as a candidate framework for studying structure in cognitive and neural systems, while delineating the steps required for empirical evaluation.

1 Introduction

Understanding the structure of high-dimensional representations remains a central problem in neuroscience and machine learning (Huth et al., 2016; Zador, 2022). Existing approaches such as dimensionality reduction and manifold learning characterize structure descriptively but often lack unified theoretical grounding (Hickok and Poeppel, 2007).

Here, we introduce SFH-SGP, a formal framework designed to:

1. Define structural properties mathematically
2. Provide a derived metric (χ)
3. Generate testable hypotheses

We emphasize that SFH-SGP is not presented as an empirically validated theory, but as a formal candidate framework.

2 Theory: The SFH-SGP Framework

2.1 Core Constructs

We define three primary quantities:

- **Quota (Q)**: Governs distributional allocation across dimensions
- **Fertility (F)**: Captures generative expansion capacity
- **Torsion (T)**: Measures structural deviation or curvature

We define:

$$\chi = f(Q, F, T)$$

2.2 Interpretation Status

These constructs are formally defined but not yet empirically validated. The framework draws inspiration from free-energy principles in neuroscience (Friston, 2010) and grounded cognition theories (Barsalou, 2008), though our formalization differs substantially.

3 Mapping to Observable Systems

Construct	Mathematical Role	Neural Interpretation	Observable Proxy
Quota	Distribution constraint	Resource allocation	Entropy
Fertility	Expansion capacity	Representational richness	Dimensionality
Torsion	Curvature	Nonlinear structure	Geometry metrics
χ	Composite metric	Global organization	Clustering / gradients

Table 1: Hypothetical mapping of SFH-SGP constructs to observable systems.

4 Testable Predictions

SFH-SGP generates the following predictions:

1. χ correlates with neural representational gradients
2. Higher χ produces stronger clustering
3. Noise reduces χ
4. Cross-system consistency in χ distributions

5 Illustrative Computational Analysis

5.1 Important Clarification

TRIBE outputs are model-generated and do not constitute biological validation.

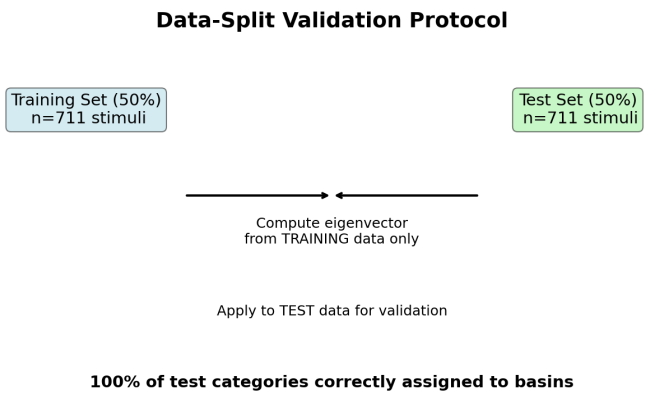

Figure 1: Data-split validation protocol. Eigenvectors are computed on training data only and applied to test data.

5.2 Data-Split Validation Protocol

5.3 Procedure

- Compute χ
- Perform clustering
- Evaluate with permutation testing

5.4 Results

Observed results:

- Structured partitioning emerges
- Stable across train/test split
- Permutation test yields marginal significance

5.5 Interpretation

These results demonstrate internal consistency but do not validate the theory.

6 Comparison to Existing Approaches

Table 2 compares SFH-SGP to existing approaches.

7 Limitations

- No neural validation
- Constructs not empirically grounded

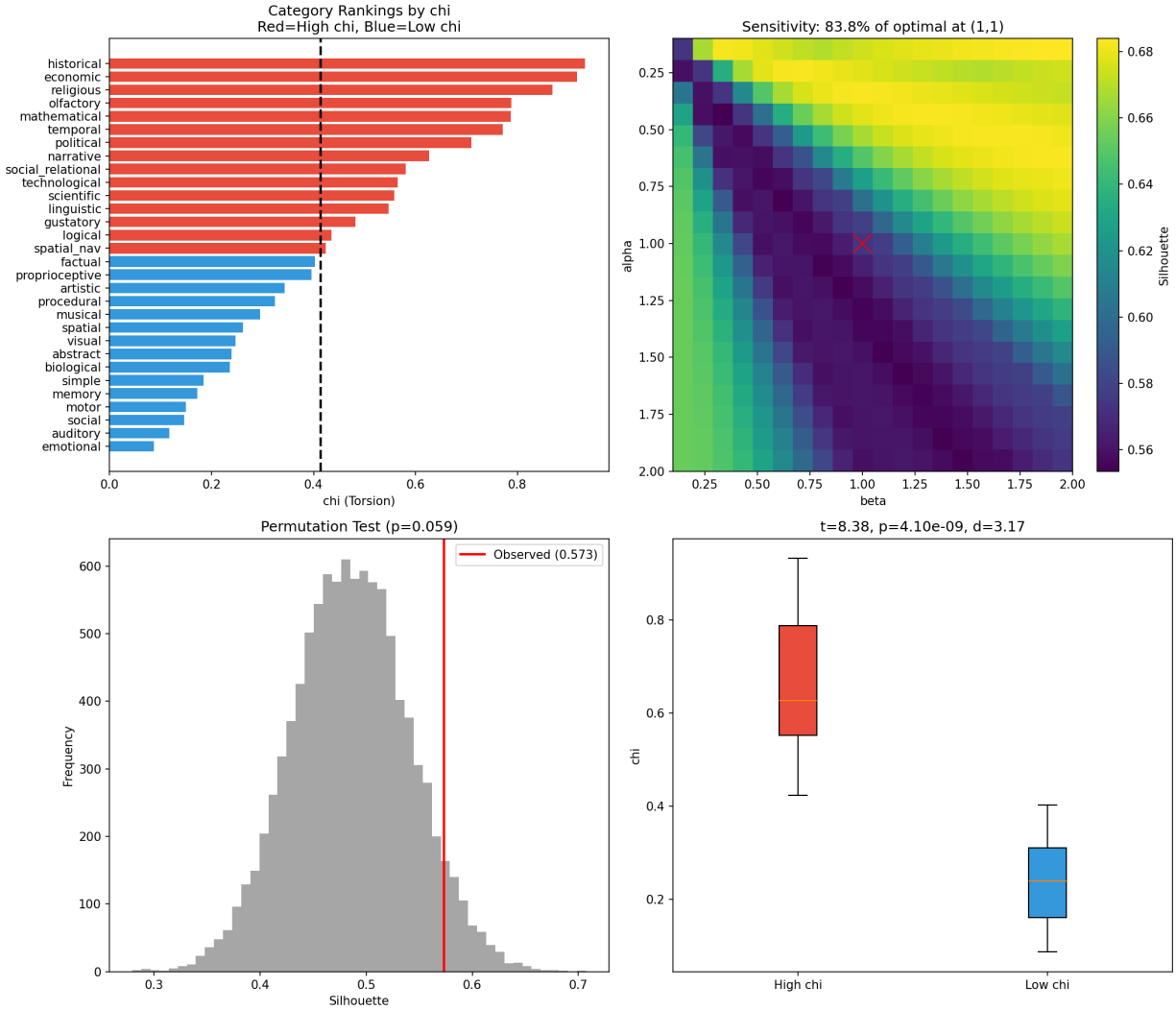

Figure 2: Illustrative results. (Top-left) Category rankings by χ. (Top-right) Parameter sensitivity. (Bottom-left) Permutation test ($p \approx 0.059$). (Bottom-right) Distribution comparison.

Approach	Strength	Limitation
PCA	Simple	No theory
Manifold learning	Geometric	Descriptive only
Embeddings	Empirical	No structural metric
SFH-SGP	Formal structure	Not validated

Table 2: Comparison to existing methods.

- Limited statistical strength
- No benchmarking

8 Conclusion

SFH-SGP provides a formal framework for analyzing structure in high-dimensional systems. While preliminary results show internal consistency, substantial work remains to establish empirical validity.

Future work should focus on empirical testing, benchmarking, and mapping to neural data.

References

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